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To cite this article: Katie Kedward, Daniela Gabor & Josh Ryan-Collins (2024) Carrots with(out) sticks: credit policy and the limits of green central banking, Review of International Political Economy, 31:5, 1593-1617, DOI: [10.1080/09692290.2024.2351838](https://doi.org/10.1080/09692290.2024.2351838)

To link to this article: <https://doi.org/10.1080/09692290.2024.2351838>



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Published online: 30 May 2024.



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Carrots with(out) sticks: credit policy and the limits of green central banking

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ABSTRACT

This article considers the role of central bank interventions in credit and financial markets in support of decarbonization. Drawing on the critical macrofinance literature, we argue that central banks are constrained in greening financial flows by their continued adherence to monetary dominance – prioritizing short-term price stability – and the structural demands of global market-based finance. This has led to a narrow focus on 'market-fixing' and 'de-risking' policy interventions, implicitly outsourcing the green transition to private finance whilst asserting that central banks cannot be seen as 'climate policymakers'. Moving beyond the constraints of this macrofinancial regime remains challenging, as we illustrate with the European Central Bank's (ECB) tilting of its corporate asset purchase program. We explore the lessons from the Post-World War II credit guidance regimes employed in many countries and consider how these could be updated to meet the challenges posed by market-based finance. Such a shift would require an evolution in the role of the central bank as a public agency, challenging current norms around independence and market neutrality.

ARTICLE HISTORY Received: 4 March 2023; Accepted: 25 March 2024

KEYWORDS

Green transition; green central banking; credit policy; credit guidance; monetary policy; critical macro finance; ECB; monetary dominance

Introduction

'We are not, and will not be, a "climate policymaker"', proclaimed Jerome Powell, the President of the US Federal Reserve in January 2023 (Powell, 2023a). The Fed would not promote a greener economy nor intervene in credit allocation – the domain of climate policymaking – but rather consider, within its regular supervisory duties, the risks of climate change only where these raised financial stability issues. In so doing, the Fed located climate policy with elected branches of government, as it was 'not the Fed's role to tell banks which businesses they can and cannot lend to'.

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Given mutually reinforcing currency and ecological hierarchies (Svartzman & Althouse, 2020), it is perhaps not a surprise that the issuer of the global reserve currency tied intimately to fossil fuels would take such a stance. More puzzlingly, European Central Bank (ECB) board member Frank Elderson followed Jerome Powell's lead in April 2023, stressing that the ECB was not, and would not be a climate policymaker either (Elderson, 2023). Yet at the time, the ECB was acting like a fully fledged one. Its measures to 'tilt' the corporate bond portfolio, initiated in 2022, created a sophisticated institutional framework for redirecting credit flows¹ from dirty to green corporate issuers. But by July 2023 the ECB had abandoned tilting, portraying it as a distraction from the primary inflation mandate. Having invested institutionally in developing a climate strategy review, a climate roadmap, and a full-fledged strategy to green both the corporate bond portfolio and collateral framework, the ECB converged to the US Federal Reserve position of 'doing little' in just two years.

The political struggles around the concept of 'climate policymaker' raise an important question. How do we study green central banking? Scholarship has mostly focused on actor preferences and behavior, seeking to explain the institutional dynamics of the climate turn in central banking, and the divergence between major central banks (Baer et al., 2021; DiLeo, 2023; Quorning, 2023; Siderius, 2023). In part, this reflects the way in which monetary scholarship in International Political Economy (IPE) prioritizes puzzles or anomalies at the expense of broader systemic issues (Cohen, 2016). What if – as Cohen calls for – we focused on the decarbonization of the economy as the broader systemic issue, and prioritized the content – i.e. the types of policies – of green central banking to address this societal challenge, alongside the institutional pathologies and international politics that constrain such policies? After all, the allocation of money and credit plays a systemic role in shaping the future direction of a capitalism dominated by global finance. Furthermore, as state institutions whose large balance sheets are increasingly entangled within the infrastructures of global market-based finance (Braun, 2020), the policy decisions of major central banks are of critical importance in disciplining carbon finance.² The architecture of policy choices is deeply political since it cuts across, as in the Powell quote above, institutional pathologies (related to central bank independence), domestic politics (green/dirty constituencies) and international negotiations in central bank forums such as the Network for Greening the Financial System (NGFS).

We make two theoretical contributions. We first analyze the institutional politics and structural pressures that have shaped green central banking. For this, we draw on critical macrofinance to distinguish between two modes of green central banking: A '*risk-based approach*' vs. '*allocative green credit policy*'. Building on work by Chenet et al. (2021) and Dafermos (2021), we articulate the dominant policy mode as a *risk-based approach* that stresses 'fixing' market signals, both by enhancing price discovery and by steering green/dirty credit and asset prices (derisking). We explain its dominance as a structural outcome of the institutional architecture of the macrofinancial status-quo, where central banks accommodate the rise of market-based finance by adjusting price signals, that is, by derisking sovereign bonds, corporate bonds, or currency, given that both monetary policy and financial stability are ultimately dependent upon the 'investability' of bond and currency markets (Gabor, 2020; Musthaq, 2021). In responding to internal and external

pressures (Deyris, 2023), central banks extend derisking to climate issues, thus preserving the institutional privileges conferred by inflation targeting. In its weak derisking form (Gabor & Braun, 2023), this involves information-based measures, such as disclosure of climate risks, or stress tests, that reinforce the epistemological and infrastructural power of private finance, allowing it to set the pace and direction of decarbonization.

However, the risk-based approach has major inconsistencies. Where it includes the decarbonization of monetary policy, in bond portfolios and collateral frameworks, it veers into the realm of credit policy that was predominant in the post-war period in many countries (Bezemer et al., 2023), but which is at odds with the claims of market neutrality at the core of inflation targeting. In particular, we show how the ECB's tilting strategy altered the scope and reach of monetary policy beyond just inflation targeting, toward sectoral allocation that involved directly monitoring and disciplining corporate financing and investment behavior, rewarding good performers and penalizing poor performers. This critical first step toward building state capacity to discipline carbon capital is vulnerable to the institutional and political frictions created by monetary dominance. Inflationary pressures, and the political leverage this gave anti-climate lobbies, saw the ECB abandon the greening of monetary policy. By unpacking such institutional politics, we argue that the risk-based approach has limited potential to discipline carbon capital and hence support decarbonization, due to its reliance on private finance-led decarbonization, the structural demands of market-based finance and its vulnerability to the reputational politics of inflation-targeting regimes that limit deviations from monetary orthodoxy (Moschella, 2022).

Instead, central banks must assume the role of 'climate policymaker' if they want to support decarbonization effectively. This would involve not just abandoning market neutrality but also a revival of credit policy – adjusted for the realities of today's market-based financial system – still a taboo in central banking circles (Monnet, 2023). Our second theoretical contribution elaborates the role of 'climate policymaker' in more detail, articulating '*allocative green credit policy*' as an alternative mode of green central banking, characterized by policymaker discretion in the selection of sectors to be (dis)incentivized for the green transition and the use of more coercive policy tools to steer credit (e.g. maximum/minimum lending targets). This draws on both the historical credit policy regimes of the 1950s and 1960s, and contemporary examples of more coercive approaches to decarbonizing finance, mainly in middle income countries.

Whilst the risk-based approach facilitates, albeit uneasily, a depoliticized, technocratic mode of green central banking, allocative green credit policy is instead grounded in more 'promotional' objectives to support broader government objectives (Baer et al., 2021). We examine the distinct institutional configurations such a mode implies, including coordination with ministries of finance, financial regulators and government agencies, complementing green fiscal and industrial policies, as well as international coordination. Finally, we reflect upon the institutional challenges and political obstacles to implementing allocative green credit policy in the context of globalized market-based finance.

The rest of the article proceeds as follows. The next section reviews the theoretical literature on green central banking and articulates critical macrofinance as our distinct theoretical lens. Next, we present the key features of the

dominant risk-based approach to decarbonizing finance and problems with 'market-fixing' and 'de-risking' policy strategies and explore the uneasy institutional politics of extending this risk-based approach to monetary policy. We then articulate an alternative policy mode to governing both bank credit and market-based finance, evaluating emerging and historical examples. Finally, we conclude by reflecting upon the institutional and political implications of our analysis.

Reading green central banking through a critical macrofinance lens

In recent years, central banks around the world have introduced a variety of climate-related initiatives (D'Orazio, 2022; Dikau & Volz, 2021a). Scholarship has sought to explain how the 'climate shift' took place in individual central banks, despite political tensions arising from stretched mandates in the absence of legislative change. This literature focuses on distinctive actor groups, pointing to the influence of internal pressures, such as activist staff (Siderius, 2023) and external pressures, such as domestic policy environments (DiLeo, 2023), the NGFS's peer group pressure (Deyris, 2023) and 'field arbitrageurs' (Quorning, 2023). Scholarship also sought to account for institutional differences in the technocratic 'green turn' (DiLeo et al., 2023). For many, differences in scope and ambition can be explained by mandates, with the central banks of emerging economies typically having wider legal scope to implement policies that support government policy (Baer et al., 2021; Dikau & Ryan-Collins, 2017). In turn, Dafermos (2021) and Chenet et al. (2021) critique the risk approach taken by large central banks, justified by (short-term) financial stability considerations alone, arguing that it will not effectively decarbonize finance nor manage the system-wide risks posed by climate change.

Another strand of literature debates whether the climate shift represents a 'paradigm shift' in central banking. DiLeo (2023) argues that the Bank of England's green policies represent a 'thermostatic shift', where policy innovations have been enabled by a supportive domestic policy environment but without substantially changing the central bank's hierarchy of goals or institutional framework. This argument is bolstered by the Bank's *volte-face* on green issues with the appointment of a conservative central bank governor to replace Mark Carney, the climate central banker. Examining the ECB, van 't Klooster (2022) argues that post-2008 developments represent a paradigm shift undertaken covertly without legislative action, enabled by cloaking green policy in the rhetoric of prudential risk management. For van 't Klooster, the ECB's 'strategic ambiguity' on climate risk allows it room to maneuver whilst avoiding overt politicization. Official central bank communications become strategic tools that are necessarily cryptic in how they shape this ambiguity.

Whilst valuable, these contributions neglect the structural pressures and entanglements with the private sector that shape central bank policies in contemporary, globalized market-based finance. In contrast, critical macro-finance starts from the systemic role that money and credit play in capitalism, a 'monetary economy of production', to focus its analytical attention on the co-evolution of private finance and macroeconomic policy institutions (see Dutta et al., 2020; Gabor, 2020). Private money is not a veil on real economic activity, whose neutrality must be preserved

by an independent central bank, but rather an engine for institutional transformation (Minsky, 1986). These institutional transformations do not occur automatically, but through political struggle and contestation, particularly when the policies required to govern new forms of money and credit overtly undermine the existing policy regime. Through a critical macrofinance lens, the growing dominance of market-based finance accompanying financial globalization (Ban & Gabor, 2016) generates new structural demands and institutional pressures on both central banks and fiscal/debt public agencies, and new forms of infrastructural power for private finance (Braun, 2020). It tests the limits of the macro-institutional architecture of the past 40 years, whereby central banks target inflation under a regime of monetary dominance, leaving fiscal authorities to be disciplined by sovereign bond markets (see Schnabel, 2021; also Moschella, 2022).

The rise of market-based finance, alternatively known as shadow banking, captures the growing importance of institutional capital in private credit and shadow/money creation (Gabor, 2020). Institutional capital – a constellation of pension funds, insurance companies, sovereign wealth funds, hedge funds or high net worth individuals and their asset managers, including private equity – extends credit through capital/bond markets, and finances credit creation through wholesale money (repo) markets and derivative markets, across borders and currencies (Michell, 2017). In the new ecosystem that circulates trillions of USD dollars across credit instruments (asset classes), banks continue to create credit and money, but their balance sheets expand to connect demand for leverage from hedge funds and other aggressive yield-chasing institutions with demand for safety from institutional capital that is not covered by the state's deposit guarantee, historically limited to ensure safety for small depositors (Pozsar, 2011). Collateral-based financial instruments reconcile the inherent tensions between demand for risk and demand for safety.

This creates two sets of deeply interconnected systemic issues for both fiscal authorities in their capacity as sovereign debt issuers, and for (inflation-targeting) central banks. Sovereign bonds become systemic collateral and are therefore exposed to cyclical liquidity pressures (Gabor, 2016). During good times, demand for sovereign collateral improves their market liquidity and lowers borrowing costs, expanding fiscal space. Shortages of sovereign collateral are addressed by the private manufacturing of 'safe' collateral *via* securitization in shadow banking. Financial instability, like the collapse of Lehman Brothers or the Covid-19 pandemic, creates liquidity spirals in private collateral markets, which can rapidly propagate to destabilize sovereign bond yields, as during the Eurozone 2010–2012 crisis, or the US Treasury market in 2020. In turn, the centrality of sovereign bonds for both institutional credit and overall financial stability confronts the inflation-targeting central bank with complex dilemmas. To stabilize private finance and support credit creation, it must intervene to purchase sovereign bonds outright from secondary markets in order to backstop market liquidity, an institutional innovation which central banks term market-maker of last resort (MMLR) (Gabor, 2016).

But MMLR or even quantitative easing do not mark a return to the Keynesian logic of fiscal dominance where central banks targeted a sovereign bond yield (interest rate) to support the overall fiscal stance or credit policy objectives more broadly. Rather, CMF scholarship reads these as 'derisking' interventions: Central

banks adjust the risk/return profile of sovereign bonds (i.e. nudging rather than directly setting bond yields) in order to preserve their *investability*, required for both financial stability and private credit creation in a market-based financial system (Gabor, 2020; Musthaq, 2021). In the most recent iteration, the US Federal Reserve's Bank Term Funding Program, introduced in the wake of the Silicon Valley Bank collapse, upended decades of conservative treatment of sovereign collateral – the Fed accepted US Treasuries at par (issuance) value, and thus increased their attractiveness as collateral for private finance. A similar logic applies to unconventional corporate bond purchases. Central banks guide bond price signals in the 'right' direction but leave the price at the *discretion* of private actors.

Derisking interventions thus allow central banks to deflect political contestation around both 'sovereign debt monetization' that blurs the line between monetary and fiscal policy, and 'credit policy' in the case of its unconventional corporate bond purchase – both of which undermine the premise of central bank independence. Interventions are focussed on addressing 'market dysfunction', a code for either price or financial instability, rather than supporting any wider government objectives. Derisking is thus not transformative in its ambitions. Faced with new challenges, the central bank deploys new instruments that simultaneously maintain the macrofinancial stability of market-based finance and preserve its privileged institutional position conferred by monetary dominance.

Importantly, derisking interventions are not confined to monetary policy. Take for instance green hydrogen, viewed by many as the 'solution' to decarbonizing hard-to-abate sectors. Regulatory derisking would introduce new rules that favor green hydrogen investors, and remove subsidies/preferable treatment for fossil producers (Gabor & Sylla, 2023). Fiscal derisking can take the form of private–public partnerships where the state moves a series of risks onto its balance sheet, compensating investors for risks that demand projections do not materialize, that future governments introduce new regulations, or local currencies depreciate. It can also take the form, as in the US Inflation Reduction Act, of tax credits for (national) producers of green hydrogen.

We deploy the CMF lens to examine the IPE of green central banking in the context of these fundamental dislocations and institutional innovations around derisking. On a spectrum, we argue, green central banking slides from enhancing price discovery to minimize financial stability risks (a 'market-fixing' supervisory agenda), to the decarbonization of monetary policy *via* direct interventions in green/dirty credit asset prices (monetary derisking), and then an explicit 'credit allocation' mode subordinated to broader government strategies for decarbonization. Through a critical macro finance (CMF) lens, central banks' position on this spectrum reflects three sets of constraints: (a) institutional commitment to monetary dominance that rejects credit policy; b) (bond) market-based credit creation that sets financing terms for dirty or green activities; and (c) pressures in the international epistemic community to narrow climate issues to the US Federal Reserve's narrow concern with financial stability.

The next section now elaborates how these three dimensions structurally confine large central banks to the dominant 'risk-based' policy mode that prioritizes derisking and defers to private sector discretion in setting the pace and direction of decarbonization.

Risk-based approach I: market-fixing strategies, *cui bono?*

The dominant approach to greening central banks emerges from the macroeconomic status quo which prioritizes 'monetary dominance' (Gabor & Braun, 2023). This can be defined as the (operational) independence of inflation-targeting central banks from fiscal authorities, alongside fiscal and industrial policy conservatism that rejects direct state intervention in credit allocation and wider financial flows (Braun & Downey, 2020), under the twin assumptions of limited fiscal capacity for public green investment and the superiority of private credit market allocation (Bezemer et al., 2023). Monetary dominance interacts with the structural demands of market-based finance to constrain central bank interventions to policies that preserve the investability of assets required for private credit creation (Gabor, 2020). Within these institutional and structural constraints, the decarbonization formula becomes *private finance leads the green transition, while the state assists*. 'Assistance' works through two pillars: Enhancing price discovery (market-fixing) under a supervisory umbrella and correcting price signals (derisking) under a monetary policy decarbonization agenda. Both ultimately delegate the pace and direction of the green transition to the private sector. Following Chenet et al. (2021) and Dafermos (2021), we term this approach the '*risk-based approach*'.

Market-fixing aims to improve the clarity and provision of climate-related information and to encourage the internalization of climate-related risks into market pricing. It reflects a *single materiality* approach: Physical or transition (climate policy) risks that are not properly priced may affect the health of private (bank) balance sheets and generate financial instability. A double materiality approach would instruct private finance to consider the impact of its credit decisions on the climate crisis (Crona et al., 2021), and require muscular central bank interventions to curtail dirty credit through penalties (see Gabor & Braun, 2023). Under single materiality, climate-related risks are deemed measurable and private actors are assumed to act on disclosures of such information in their capital allocation decisions, in turn minimizing financial stability risks. The fundamental logic is that of 'market-fixing' – the central bank creates a regulatory framework for price signals to correct toward full market equilibrium – effectively also a form of 'regulatory derisking'. Transparent risk disclosure has thus become a pillar of the green central banking agenda evolving from the Basel III regulatory framework (e.g. see NGFS, 2019). This involves forward-looking, scenario-based risk analysis or incorporating private-sector-led climate risk disclosure initiatives into supervisory expectations (e.g. ECB, 2020).

This 'lowest common denominator' approach to green central banking is exemplified by the US Federal Reserve, whose Chair Jerome Powell has repeatedly emphasized the Fed's climate responsibilities as 'narrow' and 'squarely focused on prudent and appropriate risk management' (Powell, 2023b). The polarization of US climate politics and the absence of effective domestic pressure groups had initially kept the Fed silent on climate issues, despite the growing international consensus of its materiality to central banks' mandates (DiLeo et al., 2023). Yet by late 2019, pressure from the international epistemic community, led by progress at the ECB and within the NGFS, eventually generated a 'norm cascade' that saw the Fed increasingly unable to neglect the climate risk debates (DiLeo et al., 2023). Not only were US banks operating in international jurisdictions

increasingly subjected to the climate risk supervisory expectations, but NGFS membership and influence had expanded to the point where supervisory climate risk discussions had become commonplace at international forums, such as the Financial Stability Board, the Basel Committee, and the G20. Following the presidential election of Joe Biden, a more favorable domestic policy environment provided the Fed further support in joining the NGFS in December 2020, establishing a Supervision Climate Committee and a Financial Stability Climate Committee in early 2021, and launching its own exploratory climate scenario exercise by 2022. Early on, Powell maintained his overarching commitment to market neutrality, rejecting any policy that would see the Fed *'picking one area as creditworthy and another not'* (Powell, 2020). In turn, technocrats potentially more committed to extending the Fed's role beyond narrow supervision faced significant political backlash. For instance, Sarah Bloom Raskin was forced to withdraw her nomination to the Fed, following objections by anti-climate lobbies to her comments that central banks should *'incentivise a rapid, orderly, and just transition'* (Bloom Raskin, 2021).

Deferring to seemingly objective 'science-based' measures of risk safeguards depoliticization and operational independence. Yet this 'weaker' form of the risk-based approach is unlikely to generate a transformation of the state-finance relationship in support of decarbonization. Rather, it reinforces the (infra)structural power of institutional capital (Braun, 2020) through delegating epistemic authority to private actors. This is problematic for several reasons.

First, both physical and transition climate-related financial risks are subject to fundamental uncertainty, making them challenging to quantify (Chenet et al., 2021). There are no meaningful probabilities to assign to the likely occurrence of unprecedented environmental threats, while the results of forward-looking risk assessments are subject to wide uncertainty ranges. This may explain why financial institutions are reluctant to voluntarily disclose risk exposures and then act as if these results are financially material. Five years after the launch of the Taskforce on Climate-related Financial Disclosures (TCFD), an ECB supervisory assessment found that *'banks continue to significantly underestimate the breadth and magnitude of such [climate] risks, and almost all banks (96%) have blind spots in identifying them'* (ECB, 2022b). ECB board member Frank Elderson (2022) has openly decried the state of European bank climate risk disclosures, calling them *'a lot of white noise and no real substance'*.

Second, market-fixing conveys epistemic authority to private actors – an extension of 'non-state, market-driven environmental governance' (Auld et al., 2009). The most prominent risk disclosure frameworks (e.g. TCFD, 2017) have been developed almost exclusively by coalitions of multinational firms acting through non-profit entities. Similarly, ESG ratings providers are all private sector-led: Market consolidation seeing many of the original specialized providers, such as Sustainalytics, Vivid Economics and Vigeo Eiris, purchased by large index providers, consultancies and asset managers (Azizuddin, 2021). To the extent that the risk-based approach inherently positions these disclosure frameworks as the primary mechanisms for capital reallocation, the private actors behind these initiatives arguably have attained *'de facto'* regulatory power in defining new standards for what is deemed 'green investment-worthy' (Fichtner et al., 2023; Petry et al., 2021). This further blurs the division of responsibilities between banks and regulators

when assessing the viability and feasibility of assumptions underpinning risk evaluation exercises (Smoleńska & van 't Klooster, 2022).

Such regulatory blurring between public and private spheres is vulnerable to systemic 'greenwashing', as recent market-led ESG taxonomies amply demonstrate (In & Schumacher, 2021). Private sector-led methodologies are subject to perverse incentives to underestimate risks and avoid potential adverse consequences on cost of capital (Eaglesham, 2022; Smoleńska & van 't Klooster, 2022). Criticisms of the proximity of vested interests have plagued the development of public regulatory initiatives, including the contracting of Blackrock in the development of EU prudential climate rules – a decision later lambasted by the European Ombudsman (O'Reilly, 2020). Additionally, the EU sustainable taxonomy, Europe's landmark legislation for standardizing green definitions to enable market-fixing, has suffered from the 'dilution effect' of carbon lobbies, with legislation successfully amended in July 2022 to include fossil gas and nuclear energy as taxonomy-eligible sustainable economic activities (Abnett, 2022).

A third problem with the risk-based approach refers to its potentially limited ability to effectively address market-based financing of dirty activities. The complex ecosystem of institutional capital has various obscure mechanisms to move carbon assets across opaque balance sheets to circumvent the changing relative price signals created by derisking interventions, or the disclosure requirements required by market-fixing policies. Indeed, such dynamics are now acknowledged to be at play in the oil and gas lending space across North America and Europe (Allison, 2021; Porter & Deveau, 2021). For example, the Private Equity Stakeholder Project documented that the top 10 US private equity firms have increasingly been absorbing fossil assets divested by large fossil fuel companies and investors (Seidman et al., 2022). This example illustrates that, paradoxically, while regulated financial institutions may be ostensibly divesting from fossil assets, their *indirect* support for high carbon activities, including the exploitation of fossil fuels, continues *via* their private equity allocations.

Figure 1 maps the distinctive instruments through which market-based finance extends green and dirty credit. Institutional capital can directly allocate dirty financing *via* equity (and equity ETFs/funds) and bonds (and bond ETFs/funds) or indirectly *via* hedge funds and private equity funds. These alternative asset managers in turn finance dirty activities *via* private credit markets, dirty equity/bonds or by acquiring physical fossil assets. Private credit markets – by which we refer to loans extended to companies by private equity funds – were estimated to amount to USD 1.53 trillion assets under management in 2022, of which USD 438 million was in highly liquid, cash-like securities (Aramonte & Avalos, 2021). Pension funds are the largest ultimate lender *via* private markets, contributing 70% of total commitments to private equity and private credit, alongside other institutional investors and high net worth individuals.

Furthermore, when placing cash *via* the repo market, institutional capital can also provide indirect financing to holders of dirty bonds (and thus ease financing conditions for dirty issuers), by accepting those bonds as collateral. The International Capital Markets Association has proposed to approve green labeling for repos where the cash proceeds obtained are deployed for 'green' activities, even if dirty bonds are used as collateral. Thus, institutional investors can finance Exxon bonds *via* 'green' repos, while the fungibility of cash raises doubts whether the cash

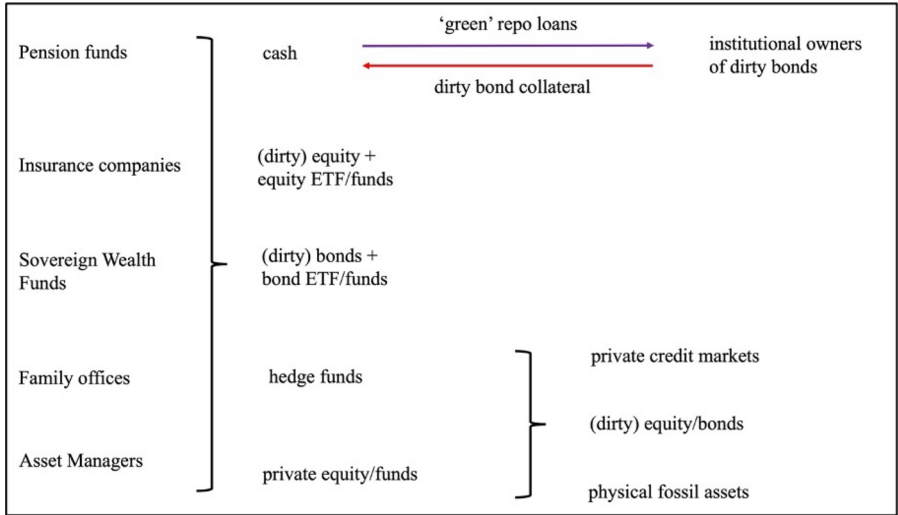


Figure 1. Avenues by which market-based finance allocates finance to dirty activities. *Source:* Authors' illustration.

proceeds from these repos can be monitored effectively or assessed to be 'green' in practice.

Taken together, the rise of market-based finance over recent decades has created multiple alternative sources of credit and capital that rival the banking sector in size and scope, but which are not subject to the same regulatory regime. Its significance lies in the potential for dirty sectors to continue to access finance even if banks successfully transition their balance sheets. Indeed, emerging reports suggest that the lack of 'ESG' oversight is at least part of the reason that oil and gas firms are increasingly willing to turn to alternative lenders over traditional bank-based financing (Porter & Deveau, 2021). Meanwhile, emerging evidence on the anti-climate voting record of large asset management firms challenges the idea that universal asset owners will act to internalize climate externalities (Baines & Hager, 2023). This in turn represents a prominent 'blind spot' in the risk-based approach to decarbonization, and a potential source of systemic financial instability. Market-fixing is hence vulnerable to the arbitrage and regulatory circumvention opportunities it creates for financial actors. The extension of green central banking from supervision to decarbonization of monetary policy attempts to address, albeit insufficiently, this blind spot.

Risk-based approach II: the uneasy institutional politics of greening monetary policy *via* derisking

The extension of climate concerns to monetary policy instruments constitutes a substantive step toward recognizing that central banks are not neutral observers of climate-private finance interactions. Instead, they are important players that shape these interactions, for instance, by sanctioning the market failure to price carbon in their treatment of corporate bonds pledged as collateral in routine monetary policy operations or unconventional bond purchases (Dafermos et al., 2021).

Early attempts to conceptualize the link between monetary policy and climate change first explored how extreme climate events generate macroeconomic volatility that can obstruct price stability and impair monetary policy implementation (Coeure, 2018). Hence, it becomes legitimate for central banks to ask if they should do more to prevent climate change in the first place, that is, to take steps for protecting the monetary transmission mechanism from climate volatility. This framing, deployed for instance by the ECB in the early days of the climate turn, overtly undermines the US Federal Reserve argument that an inflation-targeting mandate necessarily restricts climate issues to financial stability risks and a supervisory agenda. Instead, central banks' monetary choices amplify, or conversely attenuate, climate-related price instability (Coeure, 2018) and climate action falls within the primary mandate of price stability.

Once climate becomes a monetary policy issue, inflation-targeting central banks have two broad sets of options with distinctive politics and echoes of credit policy. As a 2021 NGFS report alludes to, these boil down to a 'carrots vs. sticks' question. The first is a 'carrots for green finance' logic that deploys monetary tools to mobilize green finance, that is, improving the investability of green assets by leveraging existing market mechanisms *'in order to encourage lenders to originate or invest more in low-carbon and transition assets'* (NGFS, 2021, p. 6). The second, 'carrots and sticks', simultaneously rewards climate performers and penalizes climate laggards. But this latter requires central banks to suspend the market neutrality principle at the heart of inflation targeting, and to abandon the claim that inflation-targeting central banks should not engage in credit policy. As the NGFS (2021) report put it, *'typically, a central bank does not seek to target individual firms, households, regions, or economic sectors'* and cautioned that central banks *'should refrain from using its powers to carry out tasks that... might more properly be the responsibility of governments'* (NGFS, 2021, p. 13).

Consider the policy options outlined by the NGFS (2021) in Table 1. Central banks can choose from a menu across credit operations, collateral framework and asset purchases. Most options can either be limited to derisking green assets, or derisking green assets in parallel with penalties for dirty assets.

Green derisking adjusts credit price signals toward more favorable risk/return profiles, 'mobilising' private investment capacity through an effective subsidy – 'carrots without sticks' (Gabor, 2020). Investors managing the glut of institutional capital often argue that green projects are often non-investible: Their risk/return profile is inconsistent with investor preferences or mandates. This influential narrative has become a strategic priority for the NGFS, which proposed that central banks become 'convenors, facilitators and influencers' that directly engage in derisking exercises to create 'innovative financing instruments' and 'attract a broader range of private investors' into new asset classes, working in conjunction with multilateral development banks (Menon, 2022).

Take positive screening, one of the clearest examples of green derisking (item six in Table 1). Central banks provide favorable collateral treatment to 'sustainable' collateral, de facto allowing banks to finance these credit instruments on cheaper terms than in private markets. This is a derisking intervention in that the central bank improves the risk/return profile of green assets without fixing it or without putting in place an institutional mechanism for monitoring the climate performance of the corporate issuers of those green bonds.

Table 1. NGFS policy options for climate-aligning monetary policy tools.

Type	Policy	Description
Credit operations	1. Adjust pricing to reflect counterparties' climate-related lending	Make the interest rate for central bank lending facilities conditional on the extent to which a counterparty's lending (relative to a relevant benchmark) is contributing to climate change mitigation and/or the extent to which they are decarbonizing their business model.
	2. Adjust pricing to reflect the composition of pledged collateral	Charge a lower (or higher) interest rate to counterparties that pledge a higher proportion of low-carbon (or carbon-intensive) assets as collateral or set up a credit facility (potentially at concessional rates) accessible only against low-carbon assets.
	3. Adjust counterparties' eligibility	Make access to (some) lending facilities conditional on a counterparty's disclosure of climate-related information or on its carbon-intensive/low-carbon/green investments.
Collateral framework	4. Adjust haircuts	Adjust haircuts to better account for climate-related risks. Haircuts could also be calibrated such that they go beyond what might be required from a purely risk mitigation perspective in order to incentivize the market for sustainable assets.
	5. Negative screening	Exclude otherwise eligible collateral assets, based on their issuer-level climate-related risk profile for debt securities or on the analysis of the carbon performance of underlying assets for pledged pools of loans or securitized products.
	6. Positive screening	Accept sustainable collateral so as to incentivize banks to lend or capital markets to fund projects and assets that support environmentally friendly activities (e.g. green bonds or Sustainable Development Goals-linked assets).
	7. Align collateral pools with climate-related objective	Require counterparties to pledge collateral such that it complies with a climate-related metric at an aggregate pool level.
Asset purchases	8. Tilt purchases	Skew asset purchases according to climate-related risks and/or criteria applied at the issuer or asset level.
	9. Negative screening	Exclude some assets or issuers from purchases if they fail to meet climate-related criteria.

Adapted from NGFS (2021, p. 18).

Across the other instruments, derisking includes improvements in the liquidity of green bonds, either by outright purchases or preferential collateral treatment (see Table 1), and cheaper refinancing conditions for banks' green loans. In recent years, several large central banks have announced green policy innovations that fall squarely into this category. The Bank of Japan (BoJ) became the first major central bank to launch green refinancing in 2021, allowing banks to access long-term zero-interest refinancing for green lending. Importantly, it decided to leave loan allocation decisions entirely to banks – including the interpretation of whether a loan is 'green' or not (Clark, 2022). The BoJ instead trusts that climate risk disclosure frameworks (most notably the private-sector-led TCFD) would work as the disciplining mechanisms against greenwashing (Nishimura, 2021).

Elsewhere, the ECB began in January 2021 to accept sustainability-linked bonds as eligible collateral for Eurosystem liquidity operations and for outright monetary policy purchases. Central bank collateral eligibility criteria render assets more valuable for private finance, increase demand, and ease financing conditions for the issuer. Crucially, however, only bonds with sustainability performance targets corresponding to the EU Sustainable Taxonomy regulation are accepted – ultimately a weak regulatory mechanism given its dilution by private interests (see the previous Section).

Without guardrails, green derisking leaves the amount and type of green credit created at the discretion of private actors, and in turn protects central banks from charges of sectoral credit preferences. Yet it fails to provide much of a mechanism to green the investment intentions of institutional 'shadow' lenders. Rather, the opposite is likely to be true: Higher relative interest rates for dirty credit resulting from green derisking are likely to attract these yield-hungry shadow lenders, who in many cases are not subject to the same shareholder or regulatory pressures regarding climate risk disclosure.

In contrast, the carrots and sticks approach – e.g. Negative Screening or Tilting in Table 1 – is a clear incursion toward credit policy. Take tilting – the proposition that central banks' portfolios of corporate bonds should be 'tilted' from dirty bonds to green bonds in an attempt to correct the carbon bias inherent in a monetary policy framework that follows the principle of market neutrality. This fundamental pillar of central bank independence instructs central banks to follow the existing bond market structure in their unconventional purchases of corporate bonds. If the bond market allocates half of credit instruments to dirty corporate issuers like Exxon, given its failure to price in Exxon's carbon footprint, then central banks that follow market neutrality would similarly allocate half of unconventional purchases to Exxon. Carbon-biased monetary policy perversely improves funding conditions for dirty issuers, and the market failure to price carbon more generally. Put differently, if climate laggards exacerbate the climate crisis, and amplify shocks to macroeconomic stability that in turn obstruct monetary policy, central banks that subsidize these climate laggards by following market neutrality *de facto participate* in this obstruction, undermining their own policy-making. Among large central banks, the ECB was the first to recognize the destructive logic of its carbon-biased operations (Schnabel, 2021).

Thus, extending the risk-based approach to monetary policy necessarily requires confronting market neutrality. Central banks must accept that derisking interventions need to adjust price signals according to a set of criteria that can overcome the carbon bias hardwired into market neutrality. This unavoidably repoliticises inflation targeting and sees monetary derisking veering closely toward credit policy. But it further complicates the derisking logic: As central banks reduce their purchases of dirty bonds, the price and signaling effects of titling increase funding costs for dirty issuers and may eventually threaten market liquidity. Utilizing 'sticks' threatens dirty *investability*, potentially confronting central banks with questions of financial stability in a market-based financial system to the extent that these corporate bonds are systemic (Gabor, 2020).

This uneasy tension can be traced through how the NGFS and its members developed policy positions on greening monetary policy. Take the Bank of England's proposed framework to 'green' its corporate bond purchase scheme (CBPS),

announced in November 2021. Its strategy included penalties for dirty credit that would escalate to progressively more stringent measures – tilting, asset exclusion and divestment – for issuers who do not improve their climate performance. Tilting would have worked through marginal adjustments to the *relative* prices of green/dirty assets. By avoiding sector-specific targets on prices/quantities of credit, such an approach effectively defers sectoral allocation decisions to the private sector, who are assumed to objectively evaluate climate risk information or the price signals sent by central banks and efficiently reallocate capital accordingly, under what Smoleńska and van 't Klooster (2022) term a 'deferential transition'. But the Bank took the decision to wind down its corporate bond portfolio as part of Quantitative Tightening (QT) before it had the chance to implement the escalation strategy.

By contrast, the ECB's strategy for tilting its asset purchase portfolio, and proposals to extend the approach to its collateral framework, went further in confronting the uneasy institutional politics of monetary dominance. In July 2022, the ECB announced ambitious plans to '*limit the share of assets issued by entities with a high-carbon footprint that can be pledged as collateral by individual counterparties when borrowing from the Eurosystem*' as well as '*consider climate change risks when reviewing haircuts applied to corporate bonds used as collateral*' (ECB, 2022a). Furthermore, it proposed to tilt its corporate bond holdings toward issuers with better climate performance, defined according to '*lower greenhouse gas emissions, more ambitious carbon reduction targets and better climate-related disclosures*' determined in a 'climate scorecard' (ibid., emphasis added). The climate scorecard thus creates new institutional mechanisms for closely monitoring and disciplining the climate performance of corporate climate laggards. Furthermore, this represents a significant step beyond a central bank's usual territory of monitoring bank balance sheets for financial stability purposes only, requiring it to take a discretionary view of the climate performance of corporations.

There remains a fine line to tread between risk-based sectoral allocation criteria and the outright sectoral targeting typical of credit policy. Which institutional framework should determine the discretionary choice of sectors to be favored or penalized? Whilst the high-level discussion in the 2021 NGFS report remains vague as to how this should practically be deployed, the ECB's attempt to abandon market neutrality at a time of high inflation proved to be politically challenging. Conservative lawmakers in the European Parliament tabled several amendments to paragraphs relating to the ECB's climate strategy, claiming it was a 'distraction' from its duty to tame inflation (Canepa, 2022). Such discourse of 'mission creep' was also present elsewhere in the international central banking community as inflation persisted (Rees, 2023), including by other central bank governors (Powell, 2023a; Reuters, 2023).

Under these pressures, the ECB only pursued its tilting strategy between October 2022 and June 2023, while it abandoned plans to climate-adjust collateral haircuts by December 2022, citing a lack of empirical evidence for the necessity of such a measure. It is important to note that this was a deeply political choice. The ECB could have chosen to deploy 'Ecological Tightening' instead of QT – i.e. using its climate scorecard to skew asset sales to the worst-performing issuers (Dafermos et al., 2023). This would have still had the effect of tightening monetary policy but would have simultaneously maintained its stance of neutralizing the carbon bias of its portfolio. Instead, even if it was not stated explicitly, market neutrality

superseded any logic of green discipline; the decision to deploy passive unwinding reveals monetary dominance as re-established.

The ECB's experience reveals the limits of derisking strategies applied to monetary policy in pursuit of decarbonization. Attempting to adjust the relative price signals of asset purchase portfolios directly challenges market neutrality and requires new institutional mechanisms to monitor and discipline corporate actors beyond banks – mechanisms which resuscitate the logic (if not the full ambition) of credit policy. Ultimately, the abandonment of market neutrality at a time of high inflation fuelled the already-fraught institutional politics of monetary dominance.

Overall, the risk-based regime is not likely to be effective in systematically aligning the allocation of private financial flows with the decarbonization of the economy. On the one hand, the structural demands of market-based finance and the contested institutional politics of monetary dominance constrain the actions of central banks to a narrow set of policy tools that preserve the privileged institutional position of operational independence without provoking instability within the market-based financial system upon which the functioning of monetary policy is dependent. On the other hand, the policy tools such constraints give rise to confer significant power to the private sector in determining the pace and direction of the transition, thus neglecting the dangers of regulatory capture and vested interests to obscure yet maintain profitable dirty financing through 'backdoor' financing.

Allocative green credit policy

At the current juncture, no major central bank wants to be seen as a 'climate policymaker'. Yet the inherent limitations of the risk-based approach mean that effectively decarbonizing the economy is likely to require precisely such a 'climate policymaking' central bank, using more direct credit policy measures to steer finance. What could such a credit policy look like? And which alternative macro-financial institutional configurations could address the specific challenges of reallocating capital in support of the green transition? Effectively steering finance as a 'climate policymaker' requires not just authority to deploy particular policy instruments, but also institutional configurations and mechanisms that coordinate the actions of central banks and financial authorities with other policy actors, including mechanisms to monitor progress, enforce non-compliance and resist the influences of special interest groups.

Historical uses of allocative credit policies give an indication of how such institutional forms took shape during the post-World War II period. Deployed in countries, such as England, France, Italy, South Korea and Japan to support rapid (re-) industrialization, various credit guidance tools and institutions, including publicly owned industrial development banks, were used to promote credit growth in desirable sectors and restrict it for undesirable activities (Bezemer et al., 2023; Mikheeva & Ryan-Collins, 2022).

France, for instance, developed an institutional configuration that brought together the central bank, public banks, the Treasury, other departments such as the planning ministry, under a National Credit Council (*Conseil National du Credit* – CNC) (Epstein, 2006; Loriaux et al., 1997; Monnet, 2018). This CNC set and monitored the deployment of credit policies including rediscounting, quantitative limits on refinancing credit, reserve requirements coupled with liquid asset ratios

(which required banks to hold a minimum ratio of medium-term credit relative to total assets), and credit ceilings (*encadrement du crédit*) to slow down bank lending in specific periods. The overall policy objective was to increase the creation of 'investment credit', defined as medium- (two to five years) and long-term (more than five years) loans to non-financial firms, given political consensus that a shortage of this finance was impeding capital investment and productivity growth. However, this objective was also deployed in the context of supporting broader government policy goals including increasing export credit, supporting regional development, enabling 'creative destruction' in sectors where activity was seen as too traditional (Monnet, 2018, p. 222).

The French allocative credit policy regime was largely successful in achieving its broader objectives (Loriaux et al., 1997; Monnet, 2018). Policies, such as rediscounting in particular were successful in shifting the lending pattern of commercial banks: By the early 1970s commercial banks were contributing over 30% of investment credit, up from almost zero for the majority of the 1950s period. Overall, investment credit increased from 1% of GDP in 1949 to 25% by 1973 during a period when the French economy experienced its highest rates of output growth and capital accumulation (Monnet, 2018, p. 218).

We can draw two key points from the French experience. On the one hand, the institutional arrangements needed under the French context, particularly those determining explicit coordination between fiscal and monetary authorities, have no equivalent in any major economies today. With quantitative controls utilized to control inflation rather than the modern-day price-based mechanism of raising interest rates across all sectors, monetary policy was therefore to a certain extent subordinated to credit policy objectives. This was an institutional regime of fiscal rather than monetary dominance. On the other hand, twentieth-century credit guidance was also deployed at a time when national policymakers in high and middle-income economies had tighter control over domestic financial systems, which were for the most part dominated by bank lending. Importantly, the French experience made ample use of 'sticks' as well as 'carrots', including direct limits on credit expansion. These in turn reveal two implications on the use of allocative green credit policy to support today's challenge of decarbonization. First, as argued above, effectively steering finance implies the end of monetary dominance, and a shift to distinct institutional configurations for central banks (Monnet, 2023), where inflation targeting is effectively subordinated to supporting green industrial policy goals. Second, given today's globalized market-based financial system implicates a broader range of market actors than just regulated banking institutions, green credit policy necessarily requires more policy intervention in such markets to address 'backdoor' credit creation that threatens to undermine decarbonization trajectories.

Taken together, this represents an entirely different policy mode for green central banking which we articulate as *allocative green credit policy*. What distinguishes this approach from the risk-based approach is its explicitly 'promotional' purpose to support broader government industrial policy objectives (Baer et al., 2021), its use of more coercive mechanisms to directly steer credit, and its focus on policymaker discretion to determine sector targeting. This subordination of both monetary dominance and private sector discretion to the requirements of the strategic

goals of 'Big Green State'-led transition in turn implies a shift to an entirely different macrofinancial regime, characterized by a greater degree of explicit fiscal-monetary-industrial policy coordination (Gabor & Braun, 2023).

Both the limitations of the risk-based approach and the increasingly challenging macroeconomic environment have inspired a number of proposals for policymakers to use their discretion to directly steer green capital allocation (e.g. Krebel & Van Lerven, 2022; Philipponnat, 2020; van 't Klooster & Van Tilburg, 2020). Additionally, there exist several isolated examples of emerging green central banking policies that do not fall into the category of the risk-based approach, instead taking a more direct approach to steering capital reallocation in support of decarbonization (Dikau & Ryan-Collins, 2017). For example, the Reserve Bank of India requires banks to allocate at least 40% of loans to priority sectors, including renewable energy. Similarly, Bangladesh has a minimum bank lending ratio of 15% for sustainable financing. The Brazilian central bank has previously imposed outright portfolio restrictions on certain forms of financing – such as crop expansion in the ecologically important zones (Resolution No. 3814/2009) and to borrowers who fail to comply with environmental regulations (Resolution No. 3545/2008). The People's Bank of China (PBOC) and the Chinese banking regulator (China Banking Regulatory Commission) have together deployed various tools to steer credit creation in line with Chinese government targets under the banner of their 2012 Green Credit Guidelines (GCG; Dikau & Volz, 2021b). We summarize these various policies in a typology in Table 2.

Building on existing policy proposals, we present in the right-hand column of Table 2 some examples of how allocative credit policy could be extended beyond the regulated banking sector. The price-based tools build on, for example, the work of the Financial Stability Board on a haircuts framework for reducing systemic risks in shadow banking (see Gabor, 2016); proposed portfolio restrictions for pension

Table 2. Selected allocative green credit options.

	Targeting banking system	Targeting broader institutional capital
Price-based policies	Capital requirement adjustments Reserve requirement adjustments Credit guarantees Dirty-penalizing factor for G-SIBs Large exposure limits Countercyclical capital buffer Punitive collateral haircut adjustments Interest rate floors and ceilings Subsidized credit for households/SMEs/ priority sectors (e.g. through SIBs)*	Capital requirements for allocations to dirty (alternative) assets Punitive leverage ratio Punitive collateral haircut adjustments Margin requirement adjustments
Quantity-based policies	Portfolio restrictions: outright bans financing certain sectors/assets Credit quotas Lending ratios Large-scale public investment (e.g. through SIBs) Favourable loan-to-value/debt-to-income ratios*	Outright bans on financing certain sectors/alternative assets 100% repo haircuts on dirty collateral Mandatory exclusion of dirty assets from (ESG) indexes for passive investment Ineligibility of certain assets for securitization Forced sale of dirty assets to state 'bad bank'

*Tools targeting demand for credit.

Source: Authors.

funds and insurance companies (Flood & Cumbo, 2022); and ongoing debates around simple, transparent and standardized securitization in the framework of the EU's Capital Market Union. Quantity-based credit policies could also be adapted for market-based finance. For instance, the mandatory exclusion of dirty assets from indexes marketed as ESG would set the basis for an adequate regulation of dirty investments in passive funds (Buller & Hayes, 2022), whilst a 100% haircut on securities issued by fossil fuel companies could limit the potential for 'green-washed' credit allocation *via* private repo markets. With their balance sheets sitting at the heart of market-based credit creation, central banks are arguably a key public agency in governing such a disciplining effort, having already begun to play a more active role shaping the shadow banking infrastructure for financial stability and monetary policy purposes.

A number of arguments have been put forward to explain why allocative green credit policies may be more effective than the risk-based approach in shifting financial flows to support the green transition. Evidence and theory suggest banks engage in quantity-based credit-rationing, meaning the market-determined interest rate cannot then be viewed as a reliable indicator of efficient credit allocation (Stiglitz & Weiss, 1981; Wolfson, 1996). Quantity-rationed markets are then not defined by a price equilibrium, but by quantity determination on the supply side (the bank in this case), whatever the interest rate (Werner, 2005). This literature proposes there is hence a strong case for quantitative credit guidance policies rather than relying solely on price-based signalling.

Furthermore, the evolutionary economics literature has argued that dynamic structural economic change requires proactive public policy interventions (Geels & Schot, 2007; Kattel et al., 2018). Changing price signals alone are seen as a blunt tool to stimulate green investment, because green innovation is subject to complex non-linear dynamics such as path dependency, network externalities, technological inertia and 'lock-in' effects (Hall et al., 2017). Other authors have argued that using policymaker discretion to steer finance is necessary given the wide uncertainty ranges of attempts to quantify and internalize unprecedented environmental 'risks' into market prices (Chenet et al., 2021).

Empirical evidence on the efficacy of existing green credit policies is scarce, but some emerging studies give some indication of the potential effectiveness in shifting either financing conditions or the types of sectors financed, as well as revealing potential challenges in implementation. Various analyses have shown that China's 2012 GCG have been effective in transforming highly polluting industries, if heterogeneously so (Xu & Li, 2020). For instance, Tian et al. (2022) show that the GCG was less effective for the very largest companies and those that were state-owned. Huang et al. (2023) find that the GCG's 'penalty effect' on highly polluting firms was mediated by 'the tension between environmental policies' intended consequence and the special interests embedded in the political and economic institutions typical of a large developing economy'. Elsewhere, the Brazilian central bank's experience with aligning the financing of agricultural activities with strict environmental requirements in certain ecological zones has been shown to have resulted in a material reduction in deforestation over the period 2003–2011, compared to locations not covered under the policy (Assunção et al., 2020). Whilst this study shows the effectiveness of this particular policy tool, the broader policy environment has been less supportive of curbing deforestation in recent years, with

destruction of the Brazilian Amazon substantially accelerating during the Bolsonaro government.

What these case studies demonstrate is the importance of evaluating credit policies within the broader institutional architecture within which policymakers act. Importantly, most of the examples of more direct policies discussed above remain far from being fully-fledged credit policies analogous to the French post-war credit regime. Chinese credit guidance remains one prominent exception: Its recent and present-day use of moral suasion and window guidance to direct the lending of both private and specialized public credit institutions remains closest to the French example, with China maintaining a high degree of control over its domestic financial system (Dikau & Volz, 2021a, 2021b). Elsewhere, India, Bangladesh and Brazil have central banks with clear primary mandates for price stability. Even if historically these emerging market central banks have had closer institutional links to the government, today they operate in the context of international financial subordination and pressures to adhere to operational independence, inflation-targeting and derisking in order to mitigate their external vulnerability to exchange rate and borrowing shocks (Musthaq, 2021). The membership of a large number of these emerging market central banks in the NGFS now also indicates the expansive influence of the international epistemic community in narrowing climate policy globally to a financial stability legitimization.

The *allocative green credit policy* regime therefore represents an alternative 'ideal' of green central banking that has yet to emerge in any economy in recent decades. We do not pretend to be prescriptive in how this should or could be created in practice. As was the case for historical credit guidance regimes in the twentieth century, green credit policies would have to be deployed in varying institutional configurations to reflect the distinctive institutional possibilities within each country in question (Loriaux et al., 1997; Monnet, 2018).

Conclusion

This article has utilized the theoretical insights of CMF to examine the IPE of green central banking and explain the rapid changes in attempts to deploy green policy frameworks in major central banks over recent years. No major central bank wishes to be seen as a 'climate policymaker'. Yet, at the current juncture, a 'climate policymaking' central bank may be precisely what is required to overcome the limitations of the risk-based approach in supporting the rapid realignment of financial flows required for the success of a green transition. However, a number of difficult institutional and political questions may yet impede the materialization of such a policy regime to support decarbonization.

First, new institutional forms would be needed to facilitate and legitimize deliberative coordination between central banks and other public agencies implicated in credit policy. Whilst fiscal-monetary and state-bank coordination was pragmatically deployed in the response to the Covid-19 pandemic (Ryan-Collins et al., 2023), the status quo was swiftly re-established after the pandemic shock. Exploring the issue from the Eurozone perspective, Monnet (2023) has proposed that this could take the form of a European Credit Council, a non-decision-making body of the Parliament that would assess the interactions of the ECB's decisions with broader EU financing objectives, whereby deliberation and reflexive

coordination would replace full delegation of power as an institutional form for central banking.

A second challenge concerns how to discipline market-based finance. The institutional coordination required – including pensions and insurance regulators, national securities markets authorities, and potentially extending regulatory oversight of hedge funds, private equity and ETFs – reveals the deeper challenge of how to reassert public oversight over globalized institutional capital that has hitherto remained under- (if at all) regulated – a 'wild west' of 'backdoor' private credit creation and capital allocation. Efforts to discipline and decarbonize market-based finance also intersect with the geographical imbalances that shape the IPE of financial capitalism. The asymmetric spatial distribution of the largest institutional balance sheets, which itself already exacerbates the external vulnerability of emerging capitalist economies (Bonizzi & Kaltenbrunner, 2019), inevitably means that regulators of key financial centers in the Global North will hold far more authority to shape global regulatory outcomes – a responsibility that potentially also conflicts with domestic interests in the Global North. Whilst full discussion of this topic is beyond the scope of this article, we nevertheless draw attention to the broader relevance of financial subordination (Alami et al., 2022), currency hierarchies (Svartzman & Althouse, 2020) and geographical structural power (Fichtner, 2017) in governing finance to align with decarbonization goals. These issues position international financial governance as a critical dimension of complex global environmental governance (Bernstein & Cashore, 2012).

Finally, both these challenges also point to the broader question of the political power needed to legitimize more state input in private credit allocation. Both the institutional politics of monetary dominance (Section 4), and the political power of polluting constituencies remain prominent obstacles to reinstating the logic of disciplining finance – as lobby opposition to Green New Deal policy frameworks (McGreal, 2021), as well as the European Commission's experience with anti-climate lobbies and the Sustainable Taxonomy, have revealed (Rauhala, 2022). Meanwhile, alternative coalitions with the power to drive through an allocative green credit policy mode, which would involve tackling powerful assumptions around the importance of the inflation-targeting independent central bank, are far from obvious at the current juncture. Importantly, forms of state coordination are themselves not immune to the influence of special interests, particularly where credit allocation needs may result in distributive consequences. Future scholarship on the IPE of finance should seek to further develop these questions. The current macrofinancial regime may be reaching its political and ecological limits, but alternatives are yet to take root.

Notes

1. We use the term 'credit' in this paper to refer to both bank- and non-bank credit, including the wholesale money markets and corporate bond market (see also Kothari et al., 2020).
2. Central banks have been the financial policy makers at the forefront of public discourse on green finance, since former Bank of England governor Mark Carney's seminal speech on climate-related financial risks in 2015. It is for these reasons that we focus our analysis primarily on central banks in this article. However, we also reflect upon the need for coordination with a broader array of financial regulators in Section 5.

Acknowledgments

The authors would like to thank Alex Barkawi, Olga Mikheeva, Jérôme Deyris, Kardelen Günaydin, and Matthias Täger for helpful comments on earlier drafts of this article, as well as participants at the 25th Annual Conference of the Association for Heterodox Economics (AHE) where this article was presented on the 20 June 2023. We would also like to thank the anonymous reviewers for their very constructive feedback as part of the review process.

Disclosure statement

The authors report there are no competing interests to declare.

Funding

This research was supported by the Laudes Foundation [Grant Ref: GR-071997] and Partners for a New Economy: [Grant Ref: 'Central banks, financial regulation and nature'].

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